

Gas Filled Temperature Gauge (TG-GF)

The gas filled temperature gauge measuring system is filled with pressurised inert gas as standard. A temperature change causes the internal pressure and volume in the stem to change. The deflection of the bourdon tube thus caused, is transmitted to the pointer by a movement.

A bi-metallic element which operates on the basis of coefficient of thermal expansion of metals is fitted in between the movement and bourdon tube. The bi-metallic element manufactured from two cold-welded metal strips with different thermal expansion coefficients, is used to compensate for fluctuations of ambient temperature in the case.

A comprehensive range of the standard version facilitates a variety of applications and uses. Adhering to the EN 13190 Standard, the design and construction of the gas filled temperature gauge offers a robust and reliable means of temperature measurement, with a quick response time.

The welded design system allows for best-in-class ingress protection (upto IP-68 Standard). The use of corrosion resistant material such as high grade stainless steel for both the casing and stem, makes this temperature gauge suitable for an adverse environment and with corrosive media.



Available In

Nominal sizes	4"(100 mm), 6"(150 mm), 8"(200 mm) and 10"(250 mm)
Ranges	0 to 60°C upto 600°C

Special Features

- Compliance to latest EN 13190
- Standard ranges : 0 to 60°C upto 600°C
- Overload capacity 130 % of FSD
- Ingress protection to IP68- 4" & 6" models
- Accuracy class : tolerance class 1 as per EN 13190
- Stainless steel casing
- External zero adjustment
- Remote mounting with capillary

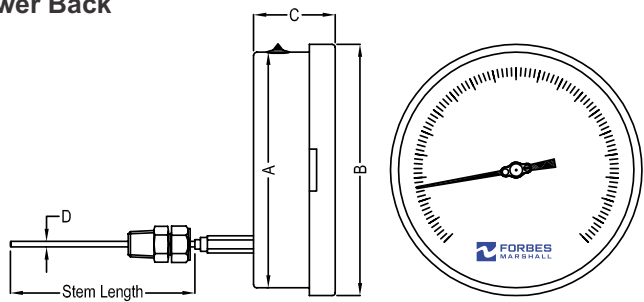
Industrial Applications

- Chemical / petrochemical industries
- Thermal, nuclear and hydroelectric power stations
- Pharmaceutical industries
- Food process industries / hygienic applications
- Water treatment plants
- Iron and steel industries
- Rubber molding / processing plants
- Hydraulics and pneumatics
- Sugar industries / fertiliser industries
- Refineries
- Cryogenics

Standard Product Specifications	
Specifications	Description
Reference standard	EN 13190
Mounting	Direct bottom entry
Connection	½” NPT (M)
Case and bezel	SS 304 (bayonet type bezel)
Stem diameter	8 mm
Stem	SS 316L
Fixed connection	SS 316
Sensing element	Inert gas filled in bulb
Dial	Aluminum, white background with black marking
Pointer	Aluminum, black color micrometer with zero adjustable
Accuracy	Class 1 as Per EN 13190 (standard)
Overload capacity	130% of maximum scale value
Window	Toughened glass
Protection	IP68 – 4” & 6” models
	IP65 – 8” & 10” Models
Gasket	Silicon
Filling plug	Neoprene
Ambient temperature	-25°C to +65°C
Process tTemperature	Max. 600°C (for dry gauge)
	Max 200°C (for silicon oil filled gauges)

Mounting Options

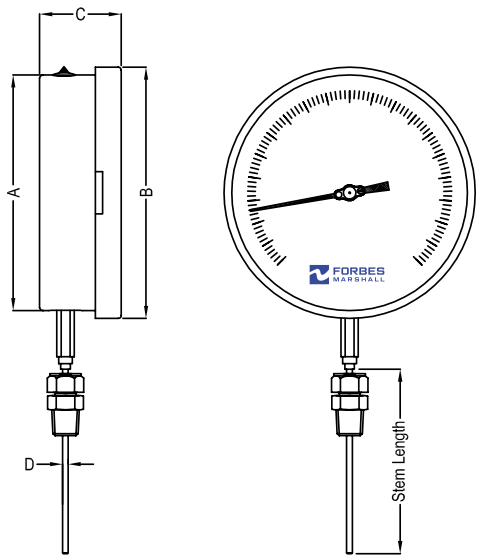
Lower Back



→ Parameters

Dial Size	A	B	C	D
100	Ø100	Ø110	48.5	6/8/9.5/10/12/14
150	Ø150	Ø160	50	6/8/9.5/10/12/14
200	Ø202	Ø215	52	6/8/9.5/10/12/14
250	Ø250	Ø264	53	6/8/9.5/10/12/14

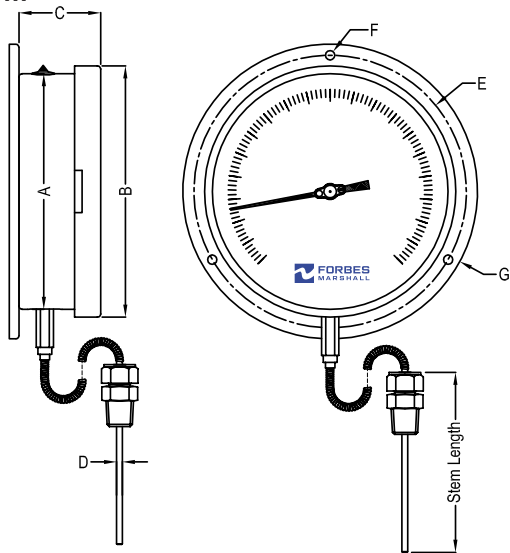
Direct Bottom



→ Parameters

Dial Size	A	B	C	D
100	Ø100	Ø110	48.5	6/8/9.5/10/12/14
150	Ø150	Ø160	50	6/8/9.5/10/12/14
200	Ø202	Ø215	52	6/8/9.5/10/12/14
250	Ø250	Ø264	53	6/8/9.5/10/12/14

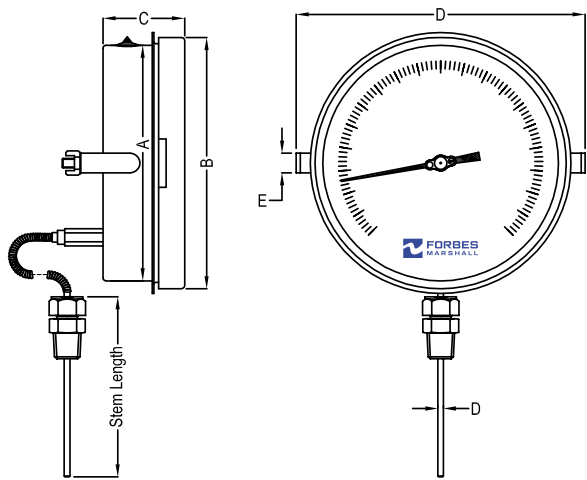
Surface Bottom



→ Parameters

Dial Size	A	B	C	G	E	F	D
100	Ø100	Ø110	48.5	Ø134	Ø118	Ø6	6/8/9.5/10/12/14
150	Ø150	Ø160	50	Ø186	Ø168	Ø6	6/8/9.5/10/12/14
200	Ø202	Ø215	52	Ø240	Ø226	Ø7	6/8/9.5/10/12/14
250	Ø250	Ø264	53	Ø290	Ø276	Ø7	6/8/9.5/10/12/14

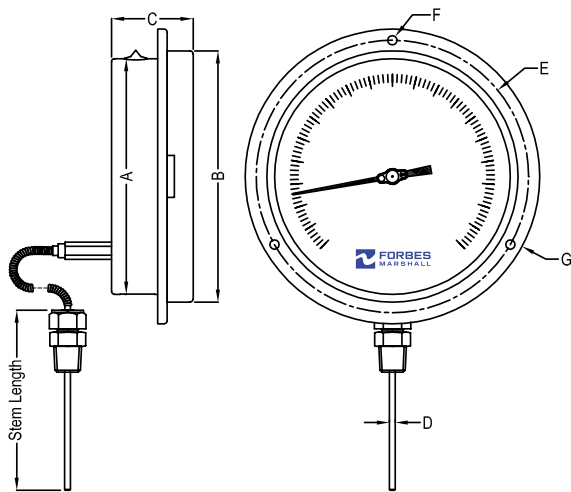
Lower Back Flush Mounting



→ Parameters

Dial Size	A	B	C	F	E	D
100	Ø100	Ø110	48.5	116	17	6/8/9.5/10/12/14
150	Ø150	Ø160	50	158.5	16	6/8/9.5/10/12/14
200	Ø202	Ø215	52	241	40	6/8/9.5/10/12/14
250	Ø250	Ø264	53	264	40	6/8/9.5/10/12/14

Lower Back Panel Mounting



→ Parameters

Dial Size	A	B	C	G	E	F	D
100	Ø100	Ø110	48.5	Ø134	Ø118	Ø6	6/8/9.5/10/12/14
150	Ø150	Ø160	50	Ø186	Ø168	Ø6	6/8/9.5/10/12/14
200	Ø202	Ø215	52	Ø240	Ø226	Ø7	6/8/9.5/10/12/14
250	Ø250	Ø264	53	Ø290	Ø276	Ø7	6/8/9.5/10/12/14

Standard Temperature Ranges

DT Range Vs Stem Diameter Vs Immersion Length (W/O Capillary)						
Code	Temperature Range (Deg-C)	Minimum Immersion Lengths				
		For Ø6 mm	For Ø8 mm	For Ø10 mm	For Ø12 mm	For Ø14 mm
A1	0 + 50	300 mm	200 mm	200 mm	150 mm	150 mm
A2	0 + 60	300 mm	200 mm	200 mm	150 mm	150 mm
A3	0 + 80	300 mm	200 mm	200 mm	150 mm	150 mm
A4	0 + 100	300 mm	200 mm	200 mm	150 mm	120 mm
A5	0 + 120	300 mm	200 mm	200 mm	150 mm	120 mm
A6	0 + 150	300 mm	200 mm	200 mm	150 mm	120 mm
A7	0 + 200	300 mm	200 mm	200 mm	150 mm	120 mm
A8	0 + 250	300 mm	200 mm	200 mm	150 mm	120 mm
A9	0 + 300	300 mm	200 mm	200 mm	150 mm	120 mm
B0	0 + 400	300 mm	200 mm	200 mm	150 mm	120 mm
B1	0 + 500	X	200 mm	200 mm	150 mm	120 mm
B2	0 + 600	X	200 mm	200 mm	150 mm	120 mm

DT Range Vs Stem Diameter Vs Immersion Length (with Capillary)					
Capillary length	Minimum Immersion Lengths (for all Ranges)				
	For Ø6 mm	For Ø8 mm	For Ø10 mm	For Ø12 mm	For Ø14 mm
1 mts – 6 mts	300 mm	200 mm	200 mm	150 mm	150 mm
7 mts – 10 mts	325 mm	250 mm	225 mm	200 mm	200 mm
11 mts – 15 mts	325 mm	300 mm	300 mm	250 mm	250 mm

General note: while selecting the range for your temperature gauge, please take into account the relative immersion length required for proper sensing.

Ordering Information

Model	
Gas filled (adjustable connection)	GA

Dial Size	
100 mm	E
150 mm	F
200 mm	G
250 mm	H

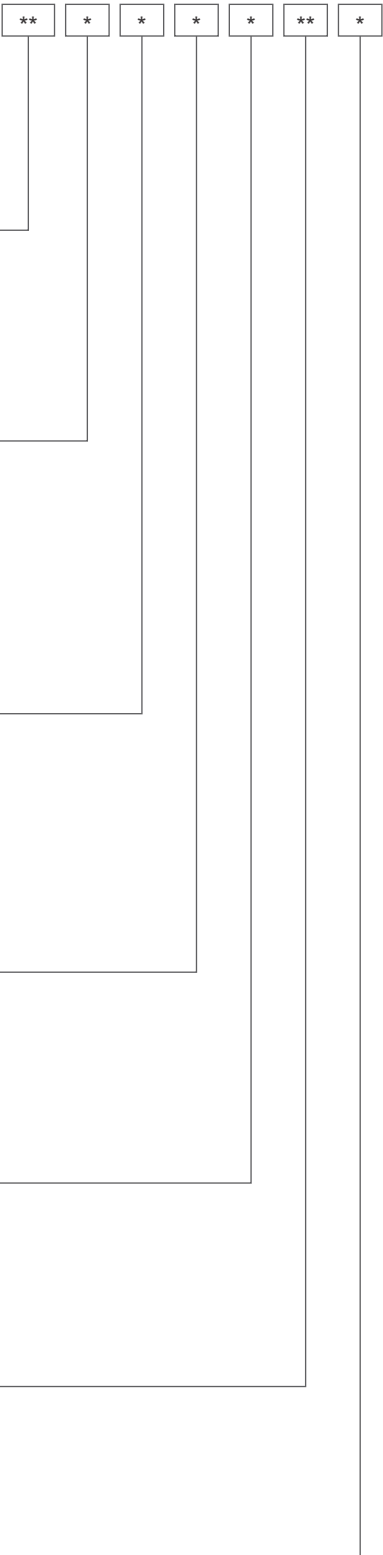
Mounting	
Every angle	0
Direct bottom	1
Lower back	2
Surface bottom	4
Lower back panel mounting	5
Lower back flush mounting	6

Connection Size	
3/8"	B
1/2"	C
3/4"	D
M20x1.5P	1

Connection Type	
NPT(M)	A
BSP(M)	B
BSPT(M)	C
Metric thread (M)	M

Range	
Refer to standard temperature range table	

Engineering Units	
Degree celsius	A
Dual scale celsius / fahrenheit	C



Configuration Generation Code

Stem Diameter	Pointer
6 mm	Micrometer Pointer
8 mm (standard)	
10 mm	
12 mm	Window
14 mm	Toughened glass (standard)
	Safety laminated shatterproof glass
Stem Length	Optional Features
300mm (standard)	130% of full scale reading (standard)
150mmTo 700mm with steps of 25mm - please specify	Dampening liquid silicon oil (upto 200°C)
	2" pipe mounting bracket in SS 304
Capillary Length	External zero adjustment
1m to15m (with PVC coating option) - please specify	Mirror scale dial for 6" dail size only
	Tag marking on dial
Case and Bezel	SS tag plate
SS 304 (standard)	Pointer stopper
SS 316	Epoxy painted case and bezel
	Custom designed dial
Bourdon Tube	Special requirement - Please specify
SS 316 (standard)	
25CRM04 (above 250°C)	Tests and Certificates
	Five point calibration certificate (standard)
Movement	NABL traceability certificate
SS 304 (standard)	Material test certificate
SS 316	EN 10204 3.1.B certificate
	Special requirement – please specify
Stem	
SS 316	
SS 316L (standard)	
Connection	
SS 316 (standard)	
SS 316L	

Ordering example: GAE1CAA5A-0000 (with standard selection)



Forbes Marshall
Krohne Marshall
Forbes Marshall Arca
Codel International
Forbes Solar
Forbes Vyncke
Forbes Marshall Steam Systems

Plot No. A-19/2 & T-4/2, I.D.A
Nacharam, Hyderabad - 500 076
Tel : +91-40-27179223, 27171909,
27170732
Fax : +91-40-27173235

Email : sales-gauges@forbesmarshall.com, ccmidc@forbesmarshall.com

© All rights reserved. Any reproduction or distribution in part or as a whole without written permission of Forbes Marshall Pvt Ltd, its associate companies or its subsidiaries ("FM Group") is prohibited.
Information, designs or specifications in this document are subject to change without notice. Responsibility for suitability, selection, installation, use, operation or maintenance of the product(s) rests solely with the purchaser and/or user. The contents of this document are presented for informational purposes only. FM Group disclaims liabilities or losses that may be incurred as a consequence of the use of this information.

B-85, Phase II, Chakan Indl Area
Sawardari, Chakan, Tal. Khed
Dist. Pune - 410 501. INDIA
Tel : 91(0)2135-393400

A-34/35, MIDC H Block
Pimpri, Pune - 411 018. INDIA.
Tel : 91(0)20-27442020, 39851199
Fax : 91(0)20-27442040

CIN No.: U28996PN1985PTC037806

www.forbesmarshall.com